

Ex. Show $\lim_{x \rightarrow 0} \sin(\frac{1}{x}) \neq 0$

Pf.

Goal: Find an $\varepsilon > 0$ s.t. the definition of limit existing doesn't hold. s.t. for all $\delta > 0$ with $0 < |x - a| < \delta$

We have $|f(x) - l| < \varepsilon$

Recall $\lim_{x \rightarrow a} f(x) = 0$

$\Leftrightarrow \forall \varepsilon > 0, \exists \delta > 0$ s.t.

whenever $0 < |x - a| < \delta$

We have that $|f(x) - l| < \varepsilon$.

We claim that $\varepsilon = \frac{1}{2}$ works

Let $\delta > 0$. We want to find some x with $0 < |x| < \delta$ with $|\sin x| > \frac{1}{2}$.

Take $N \in \mathbb{N}$ s.t.

$$\frac{1}{N} < \delta$$

Let's try to find $x \in (0, \frac{1}{N})$.

$$\text{s.t. } \sin \frac{1}{x} = 1.$$

Choose k , s.t.

$$\frac{2}{1+4k} < \frac{2}{4k} = \frac{1}{2k} < \frac{1}{N}$$

$$\boxed{\begin{aligned} X &= \frac{2}{\pi} \cdot \frac{1}{1+4k} \\ &< 2 \cdot \frac{1}{1+4k} \end{aligned}}$$

If $k = N$, then $x < \frac{1}{N}$ and $\sin \frac{1}{x} = 1$



Recall

$$f(x) = \begin{cases} 0 & \text{if } x \text{ irrational} \\ 1/q & \text{if } x = p/q \text{ in lowest terms, } q > 0 \end{cases}$$

Show $\lim_{x \rightarrow \sqrt{2}} f(x) = 0$.

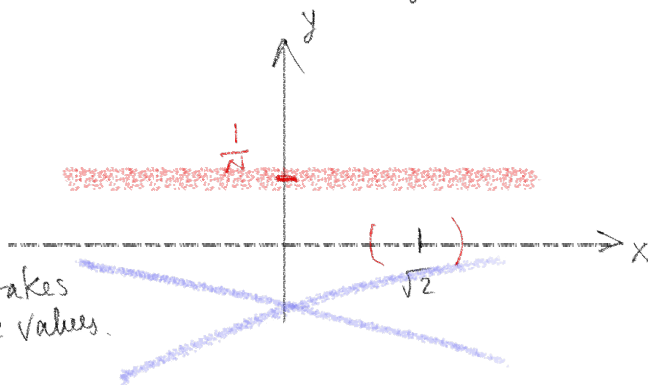
Goal For any $\varepsilon > 0$. Want to find $\delta > 0$

s.t. whenever $0 < |x - \sqrt{2}| < \delta$

We have $|f(x)| < \varepsilon$

Find $N \in \mathbb{N}$ s.t. $\frac{1}{N} < \varepsilon$.

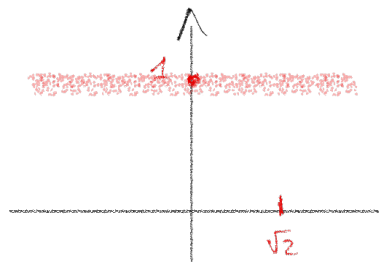
We will try to find δ assuming that $\varepsilon = \frac{1}{N}$.



Rough Work

If $N=1$.

Since the only x s.t. $f(x)=1$ are $x \in \mathbb{Z}$ (everything else gives $f(x) < 1$).



Want δ s.t. $\mathbb{Z}$ are not in $(\sqrt{2}-\delta, \sqrt{2}+\delta)$ (eg. if $\delta=0.1 \rightarrow (1.3, 1.5)$).

If $N=2$

We want to avoid $\mathbb{Z}$, but also $(-\frac{1}{2}, 0, \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots) \frac{1}{2}\mathbb{Z}$.

Take $\delta=0.001$

Pf. Take N s.t. $\frac{1}{N} < \varepsilon$.

Replace ε by $\frac{1}{N}$

Look at the set.

$$\text{pts } x \text{-axis s.t. } f(x) > \frac{1}{N} = \left\{ \frac{a}{n} : 1 \leq n \leq N, \text{ rational } \# \right\}$$

Let $S = \text{set} \cap [0, 3]$.

1. Claim that S is finite (By picture)

$$\delta < \underbrace{\min_{r \in S} \{ |r - \sqrt{2}| \}}_{\substack{\text{finite set of} \\ \# \text{'s so minimum exists.}}}$$

By construction, if $x \in (\sqrt{2}-\delta, \sqrt{2}+\delta)$.

either x irrational $\Rightarrow f(x)=0$

x rational $\Rightarrow |f(x)| < \frac{1}{N}$